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Dual fan with increased air flow

Abstract

A dual fan system includes a first fan that produces a first airflow, and a second fan that produces a second airflow. An air duct includes first and second nozzles, and first and second channels. The first channel directs the first airflow into an area between the first and second nozzles. The second channel directs the second airflow into the area between the first and second nozzles. The first and second airflows combine in the second nozzle to create an output airflow.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to dual fan system for the information handling system.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) A dual fan system includes a first fan that may produce a first airflow, and a second fan may produce a second airflow. An air duct includes first and second nozzles, and first and second channels. The first channel is in physical communication with an outlet of the first fan. The first channel may direct the first airflow into an area between the first and second nozzles. The second channel is in physical communication with an outlet of the second fan. The second channel may direct the second airflow into the area between the first and second nozzles. The first and second airflows combine together in the second nozzle to create an output airflow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

(2) FIGS. **1** and **2** are perspective views of a dual fan system for an information handling system according to at least one embodiment of the present disclosure;

(3) FIG. **3** is a cross-sectional view of a dual fan system according to at least one embodiment of the current disclosure;

(4) FIG. **4** is a cross-sectional view of a dual fan system with different air flow velocities at different regions of the dual fan system according to at least one embodiment of the current disclosure.

(5) FIG. **5** is a cross-sectional view of a dual fan system with different air flow pressures at different regions of the dual fan system according to at least one embodiment of the current disclosure; and

(6) FIG. **6** is a block diagram of a general information handling system according to an embodiment of the present disclosure.

(7) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(8) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(9) FIGS. **1** and **2** illustrate a dual fan system **100** for an information handling system according to at least one embodiment of the present disclosure. For purposes of this disclosure, an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (such as a desktop or laptop), tablet computer, mobile device (such as a personal digital assistant (PDA) or smart phone), server (such as a blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

(10) Dual fan system **100** includes fans **102** and **104** and an air duct **106**. Air duct **106** includes portions **110** and **112** and edges **114** and **116**. Portions **110** and **112** include respective openings or slits **120** and **122**. Each of side edges **114** and **116** may be a single edge or may be split into two different edges that correspond to portions **110** and **112** of air duct **106**. For example, edge **114** may be formed from an edge **130** of portion **110** and an edge **132** of portion **112**. Similarly, edge **116** may be formed from an edge **230** of portion **110** and an edge **232** of portion **112** as illustrated in FIG. **2**. Dual fan system **100** may also include heat sink **202** illustrated in FIG. **2**. Heat sink **202**

may be a tight pitch fin array. Dual fan system **100** may include additional components without varying from the scope of this disclosure.

(11) Each of fans **102** and **104** may be any suitable small profile fan, such as a piezo fan. In certain examples, fans **102** and **104** may include a high-pressure head or outlet, but the air flow may be at a low rate. In an example, the configuration of dual fan system **100** may increase the air flow rate and decrease the pressure of the air flow from fans **102** and **104**. This configuration of dual fan system **100** may enable the dual fan system to provide thermal cooling within an information handling system, such as information handling system **600** of FIG. **6**. In certain examples, heat sink **202** of dual fan system **100**, illustrated in FIG. **2**, may be placed in physical communication with a component of an information handling system, such as processor **602** or **604** in FIG. **6**. In an example, the increase of the air flow rate and the decrease of the pressure of the air flow from fans **102** and **104** may improve heat dissipation for these components via heat sink **202** of FIG. **2**.

(12) In an example, fans **102** and **104** may be securely attached to air duct **106**, such that the position of the fans are fixed within dual fan system **100**. For example, the outlet of fan **102** may be securely mounted to portion **110** of air duct **106**, and the outlet of fan **104** may be securely mounted to portion **112** of the air duct. Fans **102** and **104** may be stacked together with an air gap **140** between the fans. In an example, fans **102** and **104** may be connected to air duct **106** via their outlets but the fans may be in opposite orientations, such that substantially similar surfaces of the fans are facing each other as illustrated in FIG. **1**.

(13) FIG. **3** illustrates a cross-section of a portion of dual fan system **100** according to at least one embodiment of the current disclosure. Fan **102** includes an outlet **302** and fan **104** includes an outlet **304**. Portion **110** includes a connection section **310**, a curved section **312**, an s section **314**, a slope section **316**, and a wall **318**. Portion **112** includes a connection section **320**, a curved section **322**, an s section **324**, a slope section **326**, and a wall **328**. Dual fan system **100** may include additional components without varying from the scope of this disclosure.

(14) In an example, the different sections of portions of **110** and **112** form two air flow nozzles **340** and **342** within dual fan system **100**. Nozzle **340** includes a neck **350** and an exit **352**, and nozzle **342** includes a neck **354** and an exit **356**. As illustrated in FIG. **3**, neck **350** of nozzle **340** is narrower than exit **352** of the nozzle, such that the air flow rate or velocity increases through the nozzle as illustrated and described with respect to FIG. **4** below. Similarly, neck **354** of nozzle **342** is narrower than exit **356** of the nozzle, such that the air flow rate or velocity increases through the nozzle as illustrated and described with respect to FIG. **4** below. In certain examples, exit **354** of nozzle **340** may be located roughly in the area neck **354** of nozzle **342**.

(15) In certain examples, connection section **310** may provide a physical link between outlet **302** of fan **102** and s section **314** of air duct **106**. Also, connection section **320** may provide a physical linking of outlet **304** of fan **104** to s section **324** of air duct **106**. In an example, the air flow from fan **102** may travel through channel **360** and enter air duct **106** between nozzles **340** and **342**. Similarly, the air flow from fan **104** may travel through channel **362** and enter air duct **106** between nozzles **340** and **342**. In an example, channel **360** may have a narrow opening between curve section **312** and s section **314** and this narrow opening may increase the air flow velocity from fan **102**. Similarly, channel **362** may have a narrow opening between curve section **322** and s section **324** and this narrow opening may increase the air flow velocity from fan **104**.

(16) In an example, an air flow may be pulled through gap **140** between fans **102** and **104** in the direction of arrow **370**. This air flow may be pulled, in the direction of arrow **370**, through gap **140** and into nozzle **340** via the combination of the high pressure of the air flow from fan **102** entering air duct **106** through channel **360** and the high pressure of the air flow from fan **104** entering the air duct through channel **362**. In an example, the air flow pulled from gap **140** may increase an overall amount of air flow through air duct **106**.

(17) In certain examples, both opening **120** of portion **110** in air duct **106** and opening **122** of portion **112** may be located slightly in front of exit **352** of nozzle **340** and slightly behind neck **354**

of nozzle **342**. Based on both the location of opening **120** and the high pressure of the air flow from fan **102** entering air duct **106** through channel **360**, an air flow may be pulled through the opening in the direction of arrows **372**. Similarly, based on both the location of opening **122** and the high pressure of the air flow from fan **104** entering air duct **106** through channel **362**, an air flow may be pulled through the opening in the direction of arrows **274**. In an example, the air flows pulled through openings **120** and **122** may increase an overall amount of air flow through nozzle **342** of air duct **106** in the direction of arrow **380**.

(18) In an example, the high-pressure of the air flow from fans **102** and **104** entering air duct **106** may create a negative pressure with respect to the ambient air may be created at neck **350** of nozzle **340** and openings **120** and **122**. This negative pressure may result from gap **140** and openings **120** and **122** all being open to ambient air and as a result additional air is pulled into the airflow stream through air duct **106**. In an example, the negative pressure at openings **120** and **122** may reduce the pressure of the air flow from fans **102** and **104** entering air duct **106** generally in the area of exit **352** and neck **354**. Thus, dual fan system **100** may provide a low pressure and high volume of air flow out of air duct **106** in the direction of arrow **380**. This low pressure and high volume of air flow may provide cooling to an information handling system, such as information handling system **600** of FIG. **6**. Additionally, heat sink **202**, illustrated in FIG. **2**, may provide additional thermal cooling within the information handling system.

(19) FIG. **4** illustrates a diagram of dual fan system **100** with different air flow velocities **402**, **404**, **406**, **408**, **410**, and **412** at different regions of the dual fan system according to at least one embodiment of the current disclosure. In certain examples, the darker the shading of a region the higher velocity of the air flow in that region. As the reference number increases the air flow velocity also increases. For example, velocity level **402** is the lowest level and increases to level **404**, then level **406**, followed by level **408**, then level **410**, and level **412** is the highest velocity. Dual fan system **100** may include additional velocity level regions without varying from the scope of this disclosure.

(20) As illustrated in FIG. **4**, lowest velocity level **402** may be located within fans **102** and **104** and also within gap **140** between the fans. This lowest velocity level **402** in fans **102** and **104** may be based on the fans starting to generate air flows. In an example, gap **140** may be open to ambient air, such that the velocity level **402** within the gap is at a low level. In certain examples, as the air flows are generated within fans **102** and **104** the velocity of the air flow from the fans increases to velocity level **408**. As illustrated in FIG. **4**, the air flow velocity change from velocity level **402** to velocity level **408** may be a large increase because at the region of velocity level **408** fans **102** and **104** produce the respective air flows.

(21) In an example, as the airflow from fan **102** travels through channel **360** the velocity of the airflow may increase. For example, the exit of channel **360** may be narrower than the rest of the channel, such that the velocity may increase as the airflow from fan **102** approaches the exit. As illustrated in FIG. **4**, as the airflow from fan **102** approaches the exit of channel **360**, the velocity of the airflow increases to velocity level **410** and then to velocity level **412** as the airflow is expelled from the channel. In an example, the velocity level of the airflow may decrease the further away from the exit of channel **360** the airflow travels. For example, as the airflow from fan **102** travels from the exit of channel **360** along slop section **316** and out of dual fan system **100**, the velocity of the air flow may decrease from velocity level **412** to velocity level **410**, and then to velocity level **408**. In certain examples, additional air flow may be pulled through opening **120** to increase the amount of airflow provided from dual fan system **100**.

(22) In an example, as the airflow from fan **104** travels through channel **362** the velocity of the airflow may increase. For example, the exit of channel **362** may be narrower than the rest of the channel, such that the velocity may increase as the airflow from fan **104** approaches the exit. As illustrated in FIG. **4**, as the airflow from fan **104** approaches the exit of channel **362**, the velocity of the airflow increases to velocity level **410** and then to velocity level **412** as the airflow is expelled

from the channel. In an example, the velocity level of the airflow may decrease the further away from the exit of channel **362** the airflow travels. For example, as the airflow from fan **104** travels from the exit of channel **362** along slop section **326** and out of dual fan system **100**, the velocity of the air flow may decrease from velocity level **412** to velocity level **410**, and then to velocity level **408**. In an example, additional air flow may be pulled through opening **122** to increase the amount of airflow provided from dual fan system **100**.

(23) In certain examples, airflow may be pulled from within gap **140** between fans **102** and **104**. In these situations, the airflow may have a velocity at velocity level **402** in a region located in front of air duct **106**. When the airflow enters air duct **106**, such as through nozzles **340** and **342**, the velocity of the airflow from gap **140** may increase to velocity level **406**. In an example, as the airflow from gap **140** continues to travel through air duct **106**, the velocity of the airflow may decrease to velocity level **404**. In certain examples, the airflow from gap **140**, the airflow from opening **120**, the airflow from opening **122**, the airflow from fan **102**, and the airflow from fan **104** may combine to provide a higher volume or amount of airflow from dual fan system **100** as compared to a single fan or fans without gap **140** and/or openings **120** and **122**.

(24) FIG. **5** illustrates a diagram of dual fan system **100** with different air flow pressures **502**, **504**, **506**, **508**, and **510** at different regions of the dual fan system according to at least one embodiment of the current disclosure. In certain examples, the darker the shading of a region the higher pressure of the air flow in that region. As the reference number increases the air flow pressure also increases. For example, pressure level **502** is the lowest level and increases to level **504**, then level **506**, followed by level **508**, and then level **510** is the highest pressure. Dual fan system **100** may include additional pressure level regions without varying from the scope of this disclosure.

(25) As illustrated in FIG. **5**, lowest pressure level **502** may be located within fans **102** and **104**. This lowest pressure level **502** in fans **102** and **104** may be based on the fans starting to generate air flows. In an example, gap **140** may be open to ambient air but an airflow within the gap may be pulled within air duct **106**, such that the pressure level **504** within the gap is at a slightly higher level than within fans **102** and **104**. In certain examples, as the air flows are generated within fans **102** and **104** the pressure of the air flow from the fans increases to the highest-pressure level **510**. As illustrated in FIG. **5**, the air flow pressure change from pressure level **502** to velocity level **510** may be a large increase because at the region of pressure level **510** fans **102** and **104** produce the respective air flows. The pressure level **510** is the highest because fans **102** and **104** are designed to produce airflows with high pressure levels.

(26) In an example, as the airflow from fan **102** travels through channel **360** the pressure of the airflow may decrease. For example, the farther the airflow travels along channel **360**, the pressure may decrease. As illustrated in FIG. **5**, as the airflow from fan **102** approaches the exit of channel **360**, the pressure of the airflow decreases to pressure level **508** and then to pressure level **504** as the airflow is met with a low-pressure region at opening **120**. In an example, the pressure level at opening **120** may be pressure level **502**, which in turn may be a negative pressure with respect to the ambient air on the other side of the opening. In an example, the negative pressure may cause airflow to enter air duct **106** at opening **120**, and this mixture of airflows may cause an overall pressure of the airflow to be at pressure level **504**. As the airflow from fan **102** travels from the exit of channel **360** along slop section **316** and out of dual fan system **100**, the pressure of the air flow may be at pressure level **504**. In certain examples, additional air flow may be pulled through opening **120** to increase the amount of airflow provided from dual fan system **100** at a lower pressure level than the airflow created at fan **102**.

(27) In an example, as the airflow from fan **104** travels through channel **362** the pressure of the airflow may decrease. For example, the farther the airflow travels along channel **362**, the pressure may decrease. As illustrated in FIG. **5**, as the airflow from fan **104** approaches the exit of channel **362**, the pressure of the airflow decreases to pressure level **508** and then to pressure level **504** as the airflow is met with a low-pressure region at opening **122**. In an example, the pressure level at

opening **122** may be pressure level **502**, which in turn may be a negative pressure with respect to the ambient air on the other side of the opening. In an example, the negative pressure may cause airflow to enter air duct **106** at opening **122**, and this mixture of airflows may cause an overall pressure of the airflow to be at pressure level **504**. As the airflow from fan **104** travels from the exit of channel **362** along slop section **326** and out of dual fan system **100**, the pressure of the air flow may be at pressure level **504**. In certain examples, additional air flow may be pulled through opening **122** to increase the amount of airflow provided from dual fan system **100** at a lower pressure level than the airflow created at fan **104**.

(28) In certain examples, airflow may be pulled from within gap **140** between fans **102** and **104**. In these situations, the airflow may have a pressure at pressure level **506** in a region located in front of air duct **106**. When the airflow enters air duct **106**, such as through nozzles **340** and **342**, the pressure of the airflow from gap **140** may increase to pressure level **506**. In an example, as the airflow from gap **140** continues to travel through air duct **106**, the pressure of the airflow may decrease to velocity level **504**. In certain examples, the airflow from gap **140**, the airflow from opening **120**, the airflow from opening **122**, the airflow from fan **102**, and the airflow from fan **104** may combine to provide from dual fan system **100** an output airflow having a higher volume or amount and a lower pressure as compared to a single fan or fans without gap **140** and/or openings **120** and **122**.

(29) FIG. **6** shows a generalized embodiment of an information handling system **600** according to an embodiment of the present disclosure. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **600** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **600** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **600** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **600** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **600** can also include one or more buses operable to transmit information between the various hardware components.

(30) Information handling system **600** can include devices or modules that embody one or more of the devices or modules described below and operates to perform one or more of the methods described below. Information handling system **600** includes a processors **602** and **604**, an input/output (I/O) interface **610**, memories **620** and **625**, a graphics interface **630**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **640**, a disk controller **650**, a hard disk drive (HDD) **654**, an optical disk drive (ODD) **656**, a disk emulator **660** connected to an external solid state drive (SSD) **662**, an I/O bridge **670**, one or more add-on resources **674**, a trusted platform module (TPM) **676**, a network interface **680**, a management device **690**, and a power supply **695**. Information handling system **600** includes multiple dual fan systems **100** of FIG. **1**. In an example, one dual fan system **100** includes heat sink **202** and is in physical communication with processor **604**. Another dual fan system **100** in located within the enclosure of information handling system **600** without necessarily being in physical communication with another component of the information handling system. Processors **602** and **604**, I/O interface **610**,

memory **620**, graphics interface **630**, BIOS/UEFI module **640**, disk controller **650**, HDD **654**, ODD **656**, disk emulator **660**, SSD **662**, I/O bridge **670**, add-on resources **674**, TPM **676**, and network interface **680** operate together to provide a host environment of information handling system **600** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **600**.

(31) In the host environment, processor **602** is connected to I/O interface **610** via processor interface **606**, and processor **604** is connected to the I/O interface via processor interface **608**. Memory **620** is connected to processor **602** via a memory interface **622**. Memory **625** is connected to processor **604** via a memory interface **627**. Graphics interface **630** is connected to I/O interface **610** via a graphics interface **632** and provides a video display output **636** to a video display **634**. In a particular embodiment, information handling system **600** includes separate memories that are dedicated to each of processors **602** and **604** via separate memory interfaces. An example of memories **620** and **630** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

(32) BIOS/UEFI module **640**, disk controller **650**, and I/O bridge **670** are connected to I/O interface **610** via an I/O channel **612**. An example of I/O channel **612** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **610** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I.sup.2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **640** includes BIOS/UEFI code operable to detect resources within information handling system **600**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **640** includes code that operates to detect resources within information handling system **600**, to provide drivers for the resources, to initialize the resources, and to access the resources.

(33) Disk controller **650** includes a disk interface **652** that connects the disk controller to HDD **654**, to ODD **656**, and to disk emulator **660**. An example of disk interface **652** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **660** permits SSD **664** to be connected to information handling system **600** via an external interface **662**. An example of external interface **662** includes a USB interface, an IEEE 4394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **664** can be disposed within information handling system **600**.

(34) I/O bridge **670** includes a peripheral interface **672** that connects the I/O bridge to add-on resource **674**, to TPM **676**, and to network interface **680**. Peripheral interface **672** can be the same type of interface as I/O channel **612** or can be a different type of interface. As such, I/O bridge **670** extends the capacity of I/O channel **612** when peripheral interface **672** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **672** when they are of a different type. Add-on resource **674** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **674** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **600**, a device that is external to the information handling system, or a combination thereof.

(35) Network interface **680** represents a NIC disposed within information handling system **600**, on

a main circuit board of the information handling system, integrated onto another component such as I/O interface **610**, in another suitable location, or a combination thereof. Network interface device **680** includes network channels **682** and **684** that provide interfaces to devices that are external to information handling system **600**. In a particular embodiment, network channels **682** and **684** are of a different type than peripheral channel **672** and network interface **680** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **682** and **684** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **682** and **684** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.
(36) Management device **690** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, which operate together to provide the management environment for information handling system **600**. In particular, management device **690** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I.sup.2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **600**, such as system cooling fans and power supplies. Management device **690** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **600**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **600**.
(37) Management device **690** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **600** when the information handling system is otherwise shut down. An example of management device **690** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **690** may further include associated memory devices, logic devices, security devices, or the like, as needed, or desired.
(38) Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

Claims

1. A dual fan system comprising: a first fan to produce a first airflow; a second fan to produce a second airflow; an air duct in physical communication with both the first and second fans, the air

duct including: first and second nozzles; a first channel in physical communication with an outlet of the first fan, wherein the first channel directs the first airflow into an area between the first and second nozzles; a second channel in physical communication with an outlet of the second fan, wherein the second channel directs the second airflow into the area between the first and second nozzles, wherein the first and second airflows combine in the second nozzle to create an output airflow.

2. The dual fan system of claim 1, further comprising: a first sloped section extending away from the exit of the first channel; and a second sloped section extending away from the exit of the second channel, wherein the first and second sloped section direct the output airflow out of the dual fan system.

3. The dual fan system of claim 2, wherein the first sloped section includes a first opening at a neck of the second nozzle and the second sloped section includes a second opening at the neck of the second nozzle, wherein the first and second openings expose the first and second airflows to ambient air.

4. The dual fan system of claim 3, wherein a low-pressure region near the first opening pulls a third airflow into the air duct, and a second low pressure region near the second opening pulls a fourth airflow in the air duct.

5. The dual fan system of claim 4, wherein an addition of the third and fourth airflows increases an amount of airflow in the output airflow and decreases a pressure level of the output airflow.

6. The dual fan system of claim 1, further comprising a heat sink in physical communication with the air duct.

7. The dual fan system of claim 1, wherein a gap is located between the first and second fans, wherein the first and second airflows pull a third airflow from the gap into the air duct via the first nozzle.

8. The dual fan system of claim 7, wherein an addition of the third airflow increases an amount of airflow in the output airflow and decreases a pressure level of the output airflow.

9. The dual fan system of claim 1, wherein a neck of the second nozzle is at a same location as an exit of the first nozzle.

10. A dual fan system comprising: a first fan to produce a first airflow; a second fan to produce a second airflow; an air duct in physical communication with the first and second fans, the air duct including: first and second nozzles; a first channel in physical communication with an outlet of the first fan, wherein the first channel directs the first airflow into an area between the first and second nozzles; a second channel in physical communication with an outlet of the second fan, wherein the second channel directs the second airflow into the area between the first and second nozzles, wherein a gap is located between the first and second fans, wherein the first and second airflows pull a third airflow from the gap into the air duct via the first nozzle, wherein the first, second, and third airflows combine in the second nozzle to create an output airflow; and a heat sink in physical communication with the air duct.

11. The dual fan system of claim 10, further comprising: a first sloped section extending away from the exit of the first channel; and a second sloped section extending away from the exit of the second channel, wherein the first and second sloped section direct the output airflow out of the dual fan system.

12. The dual fan system of claim 11, wherein the first sloped section includes a first opening at a neck of the second nozzle and the second sloped section includes a second opening at the neck of the second nozzle, wherein the first and second openings expose the first and second airflows to ambient air.

13. The dual fan system of claim 12, wherein a low-pressure region near the first opening pulls a fourth airflow into the air duct, and a second low pressure region near the second opening pulls a fifth airflow in the air duct.

14. The dual fan system of claim 13, wherein an addition of the fourth and fifth airflows increases

an amount of airflow in the output airflow and decreases a pressure level of the output airflow.
15. The dual fan system of claim 10, wherein an addition of the third airflow increases an amount of airflow in the output airflow and decreases a pressure level of the output airflow.
16. The dual fan system of claim 10, wherein a neck of the second nozzle is at a same location as an exit of the first nozzle.
17. An information handling system comprising: a processor; and a dual fan system in physical communication with the processor, the dual fan system including: a first fan to produce a first airflow; a second fan to produce a second airflow; an air duct in physical communication with both the first and second fans, the air duct including: first and second nozzles; a first channel in physical communication with an outlet of the first fan, wherein the first channel directs the first airflow into an area between the first and second nozzles; a second channel in physical communication with an outlet of the second fan, wherein the second channel directs the second airflow into the area between the first and second nozzles, wherein the first and second airflows combine in the second nozzle to create an output airflow; and a heat sink in physical communication with the air duct and with the processor.
18. The information handling system of claim 17, wherein the dual fan system further includes: a first sloped section extending away from the exit of the first channel; and a second sloped section extending away from the exit of the second channel, wherein the first and second sloped section direct the output airflow out of the dual fan system.
19. The information handling system of claim 18, wherein the first sloped section includes a first opening at a neck of the second nozzle and the second sloped section includes a second opening at the neck of the second nozzle, wherein the first and second openings expose the first and second airflows to ambient air.
20. The information handling system of claim 19, wherein a low-pressure region near the first opening pulls a third airflow into the air duct, and a second low pressure region near the second opening pulls a fourth airflow in the air duct.